

How to initialize the non static fields :

a) By using reference variable :

```
package com.ravi.oop;
```

```
//ELC
public class LionDemo
{
    public static void main(String[] args)
    {
        Lion simba = new Lion();
        //Initializing the lion properties using reference variable
        simba.name = "Simba";
        simba.color = "White";
        simba.height = 4.8;
        IO.println(simba.getLionInformation());
        IO.println(simba.roar());
        simba.walk();
    }
}

//BLC
class Lion
{
    String name;
    String color;
    double height;

    public void walk()
    {
        IO.println(name+" lion is walking...");
    }

    public String roar()
    {
        return name+" lion is roaring";
    }

    public String getLionInformation()
    {
        return "[Lion name is :"+name+", Color is :"+color+", Height is :"+height+"]";
    }
}
```

b) Initializing the non static fields using parameter variable :

```
package com.ravi.oop;
```

```
//ELC
public class CarDemo
{
    public static void main(String[] args)
    {
        Car car = new Car();
        car.setCarDetails("Audi",130, "White", 2026);
        car.getCarDeatils();
    }
}

//BLC
class Car
{
    String carName;
    int carSpeed;
    String carColor;
    int carModel;

    public void setCarDetails(String name, int speed, String color, int model)
    {
        carName = name;
        carSpeed = speed;
        carColor = color;
        carModel = model;
    }

    public void getCarDeatils()
    {
        IO.println("Car name is :"+carName);
        IO.println("Car Speed is :"+carSpeed);
        IO.println("Car Color is :"+carColor);
        IO.println("Car Model is :"+carModel);
    }
}
```

Role of non-static fields while creating an object :

* As we know, non static field plays a major role because it represents **Object data**.

* In Java, Whenever we create an object, A SEPARATE COPY OF ALL THE NON STATIC FIELDS WILL BE CREATED WITH AND EVERY OBJECT.

NON STATIC FIELD = MULTIPLE COPIES WITH EACH & EVERY OBJECT

Program with Diagram :

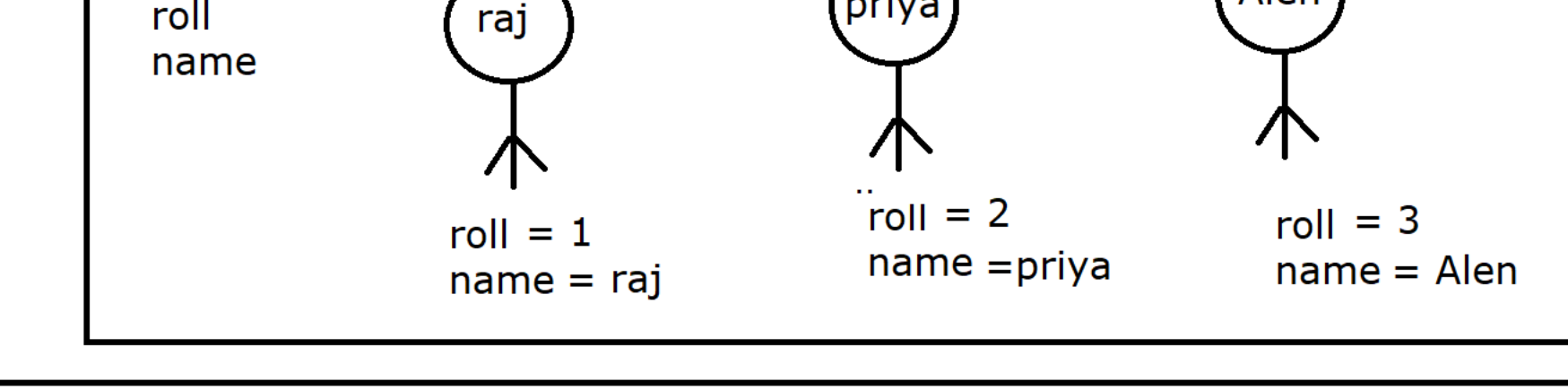
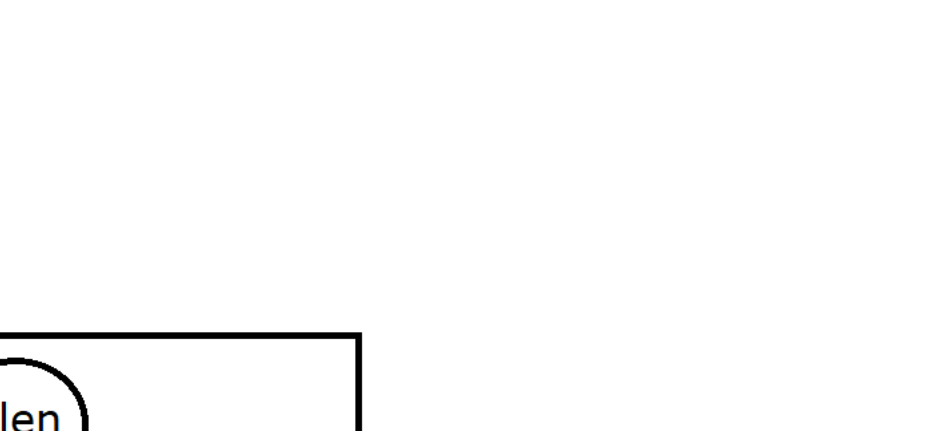
```
package com.ravi.oop;
```

```
public class NonStaticFieldCopy
{
    public static void main(String[] args)
    {
        Demo d1 = new Demo();
        Demo d2 = new Demo();
        ++d1.x;    --d2.x;

        d1.show();
        d2.show();
    }
}
```

```
class Demo
{
    int x = 100;

    public void show()
    {
        IO.println(x);
    }
}
```



Role of static field in object creation :

* For static field object is not required, any static field we can directly call with the help of class name.

* But I want to know that if, I will create an object then how many copies will be created for static field ?

* In Java, **for static field** a SINGLE COPY WILL BE CREATED and the same single copy will be sharable by all the all the objects.

Program + Diagram

```
package com.ravi.oop;
```

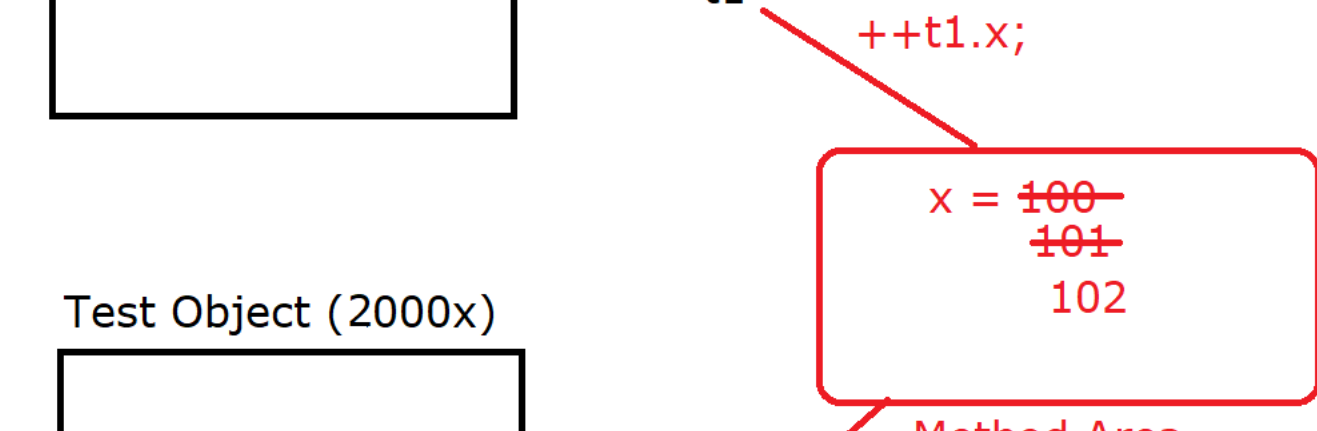
```
public class StaticFieldCopy {
    public static void main(String[] args)
    {
        Test t1 = new Test();
        Test t2 = new Test();

        ++t1.x;
        ++t2.x;

        t1.show(); //102
        t2.show(); //102
    }
}
```

```
class Test
{
    static int x = 100; //static field

    public void show()
    {
        IO.println(x);
    }
}
```



* So the conclusion is,

instance variable OR non static field = Multiple Copies with each & every Object
class variable OR Static Field = Single Copy for all the objects
That is the reason, static fields are used to SAVE THE MEMORY

Q) When we should declare a field as a non static and when to declare static ?

Non Static Field :

* If the **value of the field is different with respect to object** then we should use non static field.

Static Field :

* If the **value of the field is common with respect to all the object** then we should use static field.

Example1 :

```
public class Student
{
    int rollNumber;
    String studentName;
    String address;
    static String collegeName = "NIT";
    static String courseName = "Java Placement Batch";
}
```

Example1 :

```
public class Customer
{
    int customerId;
    String customerName;
    long accountNumber;
    static String IFSCCode = "SBIHYD010107";
    static String branchLoc = "Ameerpet";
}
```

Program on static & non static field :

```
package com.ravi.oop;
```

```
//ELC
public class StaticAndNonStatic
{
    public static void main(String[] args)
    {
        Student raj = new Student();
        raj.setStudentData(101, "Raj Gourav", "Hi-tech city");
        raj.getStudentData();

        IO.println(".....");
        Student priya = new Student();
        priya.setStudentData(102, "Priya", "Pune");
        priya.getStudentData();
    }
}

//BLC
class Student
{
    int rollNumber;
    String studentName;
    String studentAddress;
    static String collegeName = "NIT";
    static String courseName = "Adv Java Placement Batch";

    public void setStudentData(int roll, String name, String address)
    {
        rollNumber = roll;
        studentName = name;
        studentAddress = address;
    }

    public void getStudentData()
    {
        IO.println("Student Roll is :"+rollNumber);
        IO.println("Student Name is :"+studentName);
        IO.println("Student Address is :"+studentAddress);
        IO.println("College Name is :"+collegeName);
        IO.println("Course Name is :"+courseName);
    }
}
```